General Instructions for MA3923 (Applicable for all Lab Assignments)

- *Please create a folder with the last two digits of your Roll Number, IITG Email ID, the Course Number and the Lab Assignment Number. For example, if your Roll Number is "240123099", your IITG Email ID is "x.yz" and the Lab Assignment is "Lab 09" then you have to create a folder "**99x.yzMA3923lab09**".*
- *All your programs and output files (in Word/LaTeX and also in PDF format) must be put inside the folder created as per the above specification.*
- *Finally, the folder must be zipped as a .zip file and uploaded on MS Teams Group for the course, under the "Assignments" tab, within the deadline.*

All the following problems require the general linear congruential generator (LCG) given by

$$\begin{aligned} x_{i+1} &= (ax_i + b) \bmod m \\ u_{i+1} &= \frac{x_{i+1}}{m}. \end{aligned}$$

1. Generate the sequence of numbers x_i for $a = 6,\ b = 0,\ m = 11$ and x_0 ranging from 0 to 10. Also, generate the sequence of numbers x_i for $a = 3,\ b = 0,\ m = 11$, and x_0 ranging from 0 to 10. Observe the sequence of numbers generated and observe the repetition of values. Tabulate these for each group of values. How many distinct values are appearing before repetitions? Which, in your view, are the best choices and why?

2. Generate a sequence $u_i,\ i = 1, 2, \ldots, 10000$ with $m = 244944,\ a = 1597,\ b = 51749$ (choosing x_0 as per your choice). Then group the values in the ranges

   $$[0, 0.05),\ [0.05, 0.10),\ [0.10, 0.15),\ \ldots,\ [0.95, 1)$$

   and observe their frequencies (*i.e.,* the number of values falling in each group). For 5 different x_0 values, tabulate the frequencies in each case, draw the bar diagrams for these data and put in your observations. Write the choices of x_0 in the report. Further, what would be your observations if you repeat the above with (i) 1000 and (ii) 100000 generated values of u_i (instead of 10000 generated values as earlier)?

3. Generate a sequence $u_i,\ i = 1, 2, \ldots, 10000$ with $a = 1229,\ b = 1,\ m = 2048$ and choosing a value for x_0. Plot in a two-dimensional graph the points (u_{i-1}, u_i), *i.e.,* the points $(u_1, u_2), (u_2, u_3), (u_3, u_4), \ldots$. Write the choice of x_0 in the report. What are your observations?

Submission Deadline: August 01, 2026, 11:59 PM